



# Li metal-free rechargeable all-solid-state $\text{Li}_2\text{S}/\text{Si}$ battery based on $\text{Li}_7\text{P}_3\text{S}_{11}$ electrolyte

Xiaoyan Xu<sup>1,2</sup> · Jun Cheng<sup>1</sup> · Yuanyuan Li<sup>1</sup> · Xiangkun Nie<sup>1</sup> · Linna Dai<sup>1</sup> · Lijie Ci<sup>1</sup>

Received: 5 July 2019 / Revised: 15 September 2019 / Accepted: 17 September 2019 / Published online: 25 October 2019  
© Springer-Verlag GmbH Germany, part of Springer Nature 2019

## Abstract

As high energy density and enhanced safety are required for the lithium-ion battery development, all-solid-state battery has attracted significant attention. Herein, we report an all-solid-state full battery consisting of a  $\text{Li}_7\text{P}_3\text{S}_{11}$  solid electrolyte coated silicon anode,  $\text{Li}_2\text{S}/\text{graphene}$  composite cathode, and  $\text{Li}_7\text{P}_3\text{S}_{11}$  solid-state electrolyte. With a high capacity for the silicon anode and  $\text{Li}_2\text{S}$  cathode, this battery yields a high theoretical specific energy density up to  $1495 \text{ Wh kg}^{-1}$ , which is higher than that of lithium-ion batteries based on oxide cathodes. Moreover,  $\text{Li}_2\text{S}$  cathode is used as lithium source instead of using metallic lithium, avoiding interface reaction between lithium metal anode and sulfide electrolyte  $\text{Li}_7\text{P}_3\text{S}_{11}$ . This all-solid-state battery system we proposed could avoid the safety issues associated with the use of lithium metal and be a promising candidate for an enhanced energy density, which would promote their applications in the fields of electric vehicles and portable electronics.

**Keywords** All-solid-state batteries ·  $\text{Li}_7\text{P}_3\text{S}_{11}$  electrolyte ·  $\text{Li}_2\text{S}$  cathode, Si anode · Interfacial compatibility

## Introduction

The all-solid-state battery is considered as one of the most promising sustainable technology for the future energy storage and transportation with enhanced safety and higher energy density [1–4]. Solid electrolytes used in lithium-ion batteries mainly contain lithium-ion-conductive solid polymers, ceramics, and organic–inorganic composites. Basically speaking, the polymer electrolytes exhibit better interface compatibility with electrode, as the inorganic solid electrolyte possess higher ionic conductivity at room temperature [5–7].

Considering the merits of polymer electrolyte and inorganic solid electrolyte, the hybrid solid electrolyte attracts increasing interest owing to improved ionic conductivity and

interface compatibility. It has been reported that  $\text{Li}_6\text{PS}_5\text{Cl}/\text{PEO}$  and  $\text{LLZO}/\text{PEO}$  composite solid electrolytes show enhanced mechanical property and stable lithium/electrolyte interface [8, 9]. With the same issues for the all-solid-state sodium batteries, some novel electrolyte such as composite solid electrolyte or poly (ionic liquid) are used to improve the cell performance [10, 11]. In addition to ionic conductivity and interface stability, rational design of the composite electrode and preparation of the all-solid-state cells are important factors to realize high-performance all-solid-state batteries [12–14].

The Li–S battery has undergone a resurgence of interest from researchers in recent years as it yields a high theoretical specific energy density up to  $\sim 2600 \text{ Wh kg}^{-1}$  [15, 16]. Moreover, the solid electrolyte could successfully forbid the “shuttle effect” for sulfur cathode and eliminate the safety hazards associated with flammable organic liquid electrolytes [17–19]. Therefore, all-solid-state Li–S battery has the potential to overcome the capacity limitation and safety problems for mobile applications. Given that some electrolytes (such as sulfide-based solid electrolytes) already have a great ionic conductivity as high as  $10^{-3}$  or even  $10^{-2} \text{ S/cm}$  at room temperature, issues of interfacial compatibility between solid electrolytes and electrodes (especially between solid electrolytes and lithium metal anode) have become the bottleneck for the development of all-solid-state Li–S battery [20–23].

**Electronic supplementary material** The online version of this article (<https://doi.org/10.1007/s10008-019-04409-z>) contains supplementary material, which is available to authorized users.

✉ Lijie Ci  
lci@sdu.edu.cn

<sup>1</sup> SDU & Rice Joint Center for Carbon Nanomaterials, Key Laboratory for Liquid-Solid Structural Evolution & Processing of Materials (Ministry of Education), School of Materials Science and Engineering, Shandong University, Jinan 250061, China

<sup>2</sup> School of Science, Shandong Jiaotong University, Jinan 250357, China

This interfacial compatibility of all-solid-state battery is largely associated with maintaining chemical and mechanical stability of the interface between electrodes and electrolyte during battery operation. However, as far as we know, lithium metal anode tends to react with almost all of the inorganic solid electrolytes that possesses high ionic conductivity at room temperature, causing unstable interfaces [24–27]. Because of highly chemical and electrochemical activity of lithium metal, reaction between lithium and solid electrolyte would easily create a  $\text{Li}^+$  depletion area, causing large interfacial resistance for the all-solid-state cells. Researches have put great effort to alleviate the interface problems, for example, utilizing Li–In alloy as anode to replace Li metal for Li–S batteries [28, 29]. Buffer layer was also used to suppress reaction between solid state electrolyte and lithium anode. A PEO polymer layer was coated on LAGP pellet surface by spin coating method to improve the interface stability of lithium metal and LAGP solid electrolyte [30]. Besides, a LPS-PEO composite solid electrolyte was applied to successfully alleviate the reaction between lithium anode and LPS [31]. Another way to overcome these interfacial issues in the Li–S system is to use a high-capacity anode material other than lithium metal and substitute sulfur with its lithiated counterpart as the cathode, which would both possess enhanced safety and high specific energy density.

Here, we propose a novel all-solid-state full battery consisting of silicon (Si) anode and  $\text{Li}_2\text{S}$  cathode.  $\text{Li}_7\text{P}_3\text{S}_{11}$  (LPS) is chosen as electrolyte due to its high ionic conductivity ( $10^{-3}$  S/cm at room temperature), wide electrochemical window ( $\sim 5$  V), and good mechanical performance. With a theoretical capacity of  $\sim 3579$  mAh  $\text{g}^{-1}$ , Si exhibits a great electrochemical performance in all-solid-state battery.  $\text{Li}_2\text{S}$  cathode is used as lithium source, avoiding reaction between lithium metal anode and the sulfide electrolyte. This new all-solid-state Li–S battery shows a promising electrochemical cycling performance.

## Experimental

### Materials synthesis

The LPS solid electrolyte was synthesized by a liquid-phase reaction method as described in our previous report using the precursors of  $\text{Li}_2\text{S}$  (Alfa Aesar) and  $\text{P}_2\text{S}_5$  (Aladdin) (the mole ratio of  $\text{Li}_2\text{S}$  and  $\text{P}_2\text{S}_5$  is 7:3) [31]. The Si@LPS was prepared using nano silicon,  $\text{Li}_2\text{S}$ , and  $\text{P}_2\text{S}_5$  as the starting materials in acetonitrile (ACN) solution. After  $\text{Li}_2\text{S}$  and  $\text{P}_2\text{S}_5$  (7:3 mol ratio) were grounded into homogeneous powder in an agate mortar, the powder and nano Si (1:1 mass ratio) were added into ACN solvent. Then, it stirred for 24 h at 50 °C and the mixture was dried at 80 °C for 12 h to remove the ACN solvent. Finally, the powder was heat-treated at 250 °C for 1 h and the Si@LPS was obtained.

The  $\text{Li}_2\text{S}$  powder (Alfa Aesar) was dissolved in the anhydrous ethanol (99.5%, Sigma Aldrich) in the glovebox to get a 0.5 M  $\text{Li}_2\text{S}$  solution. The graphene was added in the solution with a mass ratio of 10% and stirring for 12 h. Then, it was dried at 150 °C for 12 h and annealed at 400 °C for 4 h in Ar ambient to obtain the  $\text{Li}_2\text{S}/\text{G}$  composites.

### Material characterization

A Miniflex 600 instrument was employed to test the crystalline phase of the samples with Cu Ka radiation between 10 and 80°. The morphologies and microstructures of these samples were observed on a field emission scanning electron microscope (FE-SEM, HitachiSU-70) at an accelerating voltage of 10 kV.

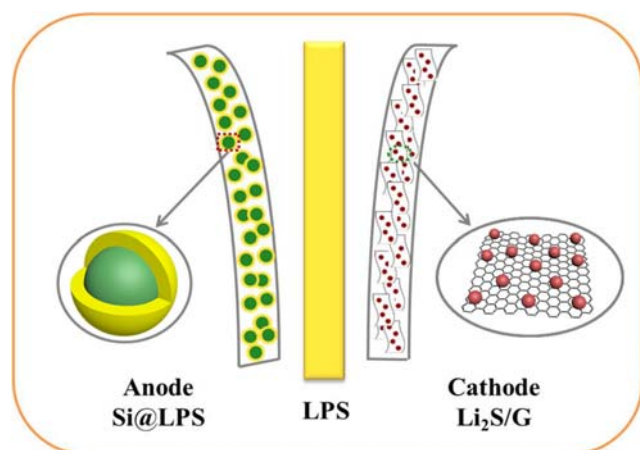
### Electrode fabrication and electrochemical measurements

For assembly of the all solid-state half batteries, the composite cathode (or anode composites) and solid electrolytes powders were sequentially put in a 10-mm die and then cold pressed at a pressure of 300 MPa. The lithium foil (75  $\mu\text{m}$ ) was attached on the other side of the solid electrolyte surface as the counter electrode and it was pressed under 120 MPa. For assembly of the all-solid-state full batteries, the composite cathode, solid electrolytes powders, and anode composites were sequentially put in a 10-mm die and then cold pressed at a pressure of 300 MPa. Cyclic voltammogram was measured on the Chenhua analyzer (chi660e) at a scan rate of 0.1 mV  $\text{s}^{-1}$  between 1 and 3 V. The galvanostatic charge/discharge test of the all-solid-state cells was carried out on a Land battery testing instrument. The cell with each experimental condition was replicated 6 times and each data is the best of individual measurements.

## Results and discussion

Figure 1 is the schematic illustration of the structure of  $\text{Li}_2\text{S}/\text{Si}$  all-solid-state full battery. The sulfide LPS is used as the electrolyte for the all-solid-state battery. A core-shell structure of LPS-encapsulated nano silicon (Si@LPS) is used as the anode. The LPS layer with 5–10 nm is coated on Si by a liquid-phase reaction method. The core-shell structure provides an intimate interfacial contact between active material Si and solid electrolyte LPS, which could promote  $\text{Li}^+$  diffusion during the lithiation and delithiation process due to the high conductivity of LPS at room temperature [31, 32]. Furthermore, our previous work reveals that the solid electrolyte LPS coating can successfully suppress Si volume expansion during electrochemical cycling [32].

The cathode is the nanocomposite of  $\text{Li}_2\text{S}$  and graphene. As  $\text{Li}_2\text{S}$  is both naturally electronically and ionically insulating,

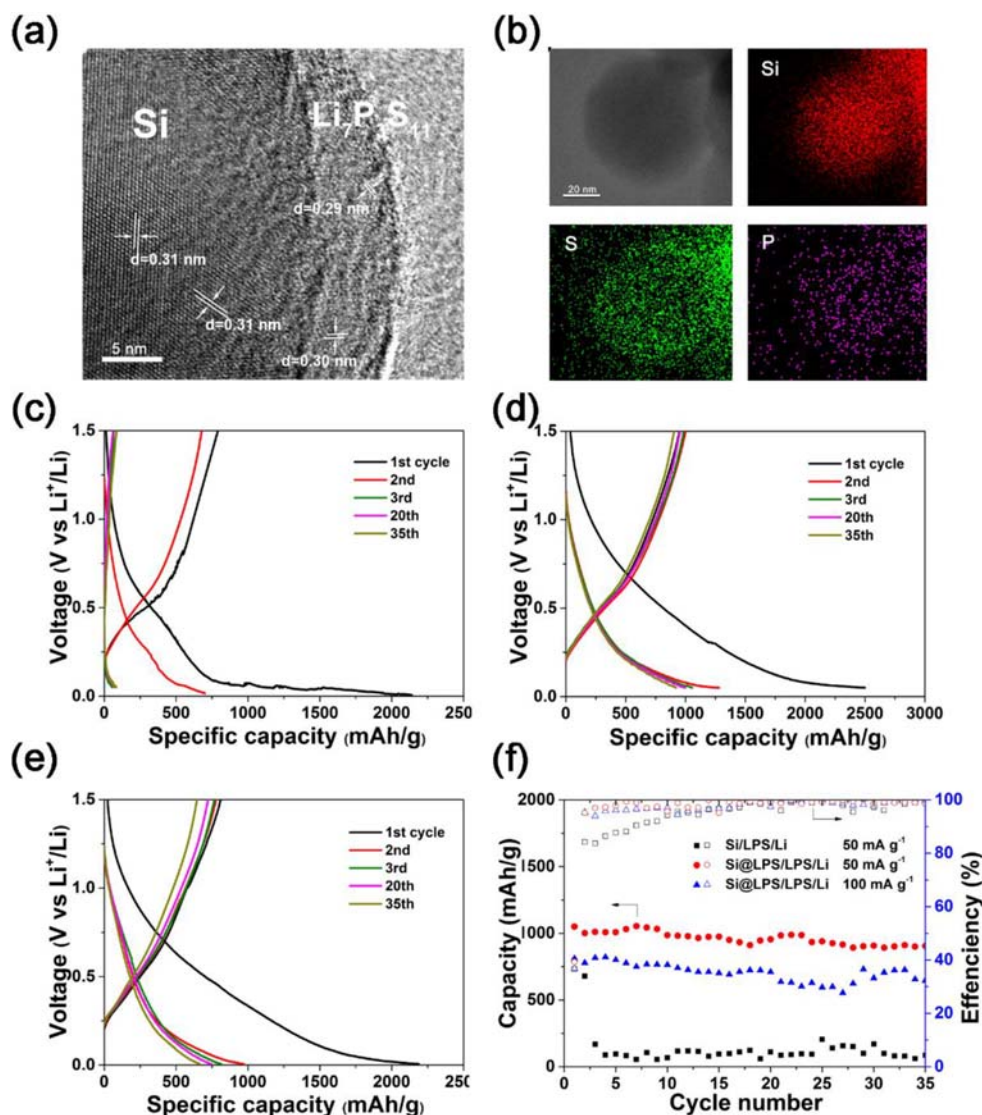


**Fig. 1** Schematic of the structure of the all-solid-state  $\text{Li}_2\text{S}/\text{Si}$  battery with LPS solid electrolyte; the cathode consists of  $\text{Li}_2\text{S}$  and graphene, and the anode contains silicon coated with LPS solid electrolyte layer

low utilization of active materials results in poor electrochemical performance when the commercial  $\text{Li}_2\text{S}$  particle is as large as several micrometers [34–37]. Decreasing the  $\text{Li}_2\text{S}$  particle size and improving the contact between  $\text{Li}_2\text{S}$  and electronic conductive additives are effective ways to resolve the problems. Graphene was used to improve electronic conductivity and physical connection between the individual  $\text{Li}_2\text{S}$  particles. Nanosized  $\text{Li}_2\text{S}$  on the surface of graphene ( $\text{Li}_2\text{S}/\text{G}$ ) composites could be obtained by dissolving commercial  $\text{Li}_2\text{S}$  powder and graphene in anhydrous ethanol and annealing at  $400^\circ\text{C}$  in Ar ambient. With improved electronic conductivity from graphene, nanosized  $\text{Li}_2\text{S}$  delivers an enhanced electrochemical performance in the all-solid-state battery.

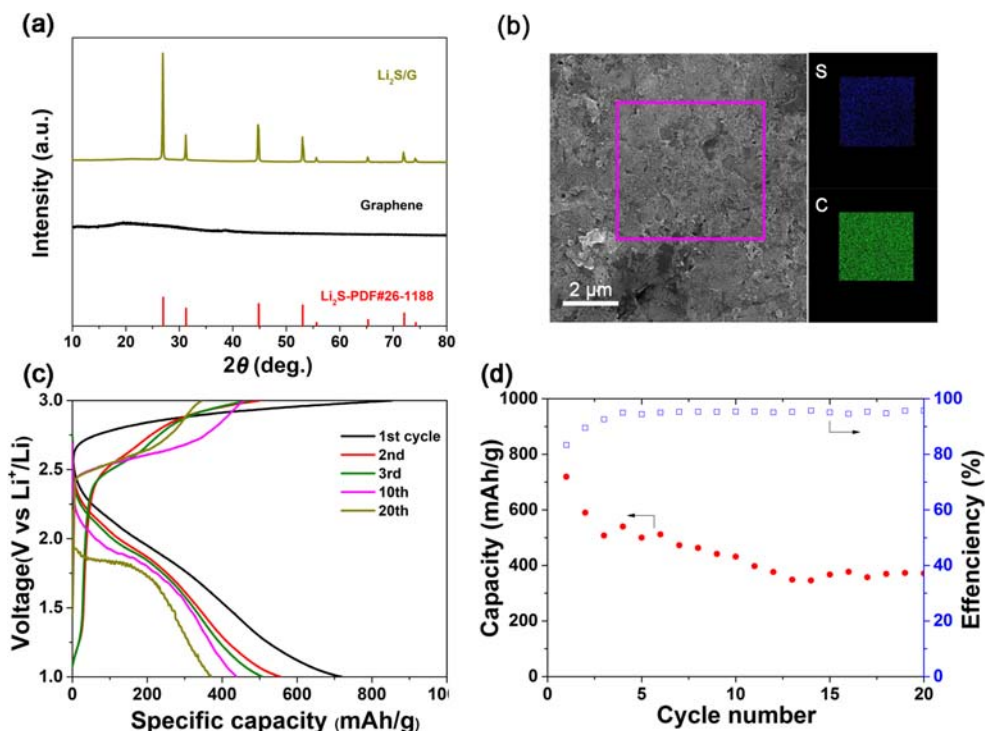
The composition and microstructure of Si and Si@LPS particles were investigated in our previous report [32]. The solid electrolyte LPS was successfully coated on the Si surface with a thickness of 5–10 nm (Fig. 2a). The STEM-EDS elemental mapping clearly shows that all elements of LPS are

**Fig. 2** **a** HRTEM image of Si@LPS. **b** STEM EDS elemental mapping images of Si@LPS, marked by the rectangle region, for Si, S, and P. The charge/discharge voltage profiles of all solid-state half-batteries based on **c** Si and **d** Si@LPS at a rate of 50 mA/g at room temperature. **e** The charge/discharge voltage profiles of all solid-state half-batteries based on Si@LPS at a rate of 100 mA/g at room temperature. **f** Cycling performances of Si and Si@LPS at a rate of 50 mA/g, and Si@LPS at a rate of 100 mA/g at room temperature





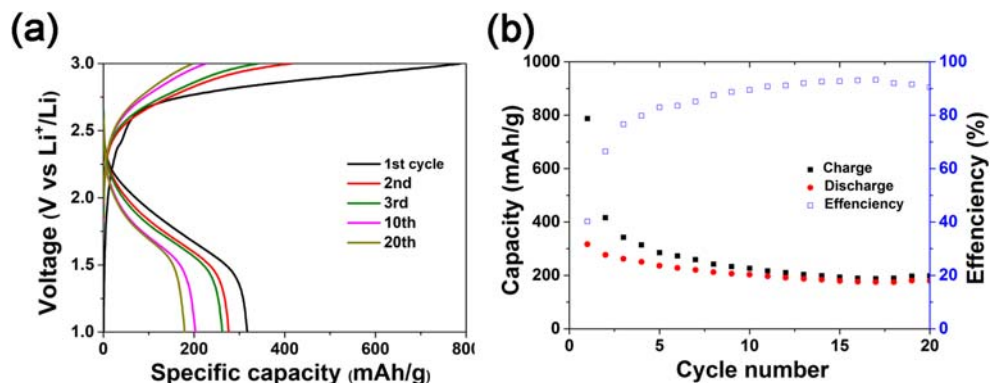
**Fig. 3** **a** XRD patterns, **b** SEM-EDS elemental mapping images of  $\text{Li}_2\text{S}/\text{G}$  composites, marked by the rectangle region, for S and C. **c** The charge/discharge voltage profiles and **d** cycling performances of all-solid-state half-batteries based on  $\text{Li}_2\text{S}/\text{G}$  at a rate of 50 mA/g at room temperature

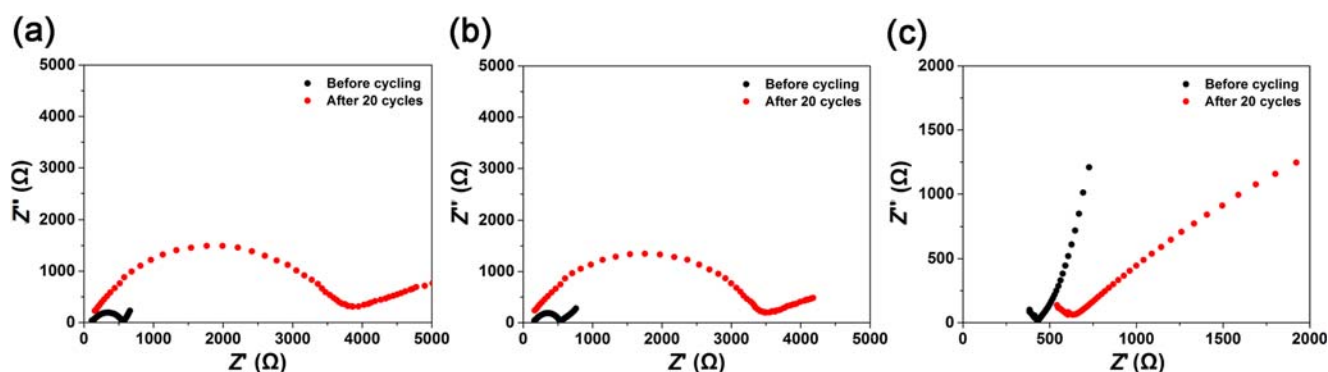


homogeneously distributed throughout the whole particle (Fig. 2b). The core-shell structured  $\text{Si}@\text{LPS}$  nanocomposite has much improved electrochemical performance as LPS electrolyte coating could promote  $\text{Li}^+$  diffusion, thus decreasing the inner resistance of the cell. The electrochemical performances of Si and  $\text{Si}@\text{LPS}$  were investigated using laboratory-scale all-solid-state half lithium cells, whereas metallic Li was used as anode. As shown in Fig. 2c–f, both curves demonstrate that the platforms of charge and discharge (0.05–0.5 V) are higher than that of conventional liquid battery, which probably results from the large polarization potential caused by the lower ionic conductivity of the all-solid-state battery [38, 39]. The first charge capacities of the Si and  $\text{Si}@\text{LPS}$  electrodes are 796.2 and 1049.9 mAh/g at a rate of 50 mA/g, respectively. The all-solid-state battery with bare Si anode shows fast capacity decay, while the cell based on  $\text{Si}@\text{LPS}$  anode exhibits much better cycling stability as displayed in Fig. 2f. After 35 cycles, the

charge capacity of  $\text{Si}@\text{LPS}$  is 904.8 mAh/g with capacity retention of 86.2%, which is much higher than that of bare Si with specific capacity of 85.9 mAh/g and 10.8% capacity retention. Furthermore, the cell with  $\text{Si}@\text{LPS}$  anode also exhibits better electrochemical performances even at a higher rate (100 mA/g) with the specific capacity of 645.8 mAh/g after 35 cycles, as shown in Fig. 2e, f. The cell with each experimental condition was replicated 6 times and it shows the experiment has good reproducibility. The improved electrochemical performance of  $\text{Si}@\text{LPS}$  anode for all-solid-state cell could be attributed to the intimate interfacial contact between the active material Si and the sulfide electrolyte, which could facilitate  $\text{Li}^+$  diffusion during charging and discharging process, and decrease interfacial charge-transfer resistance of the cell [33, 40]. Furthermore, LPS coating could suppress the volume change of Si during lithiation and delithiation process, improving the cycling stability of the batteries [32].

**Fig. 4** Voltage profile (a) and specific discharge capacity with cycling (b) of all-solid-state full battery with a  $\text{Li}_2\text{S}/\text{G}$  composite cathode and  $\text{Si}@\text{LPS}$  anode at a current rate of 50 mA  $\text{g}^{-1}$  between 1 and 3 V. The specific capacity values are calculated based on the mass of  $\text{Li}_2\text{S}$





**Fig. 5** Nyquist plots from the electrochemical impedance spectroscopy (EIS) measurements for the all-solid-state batteries of **a** Si/Li, **b**  $\text{Li}_2\text{S}/\text{Li}$ , and **c**  $\text{Li}_2\text{S}/\text{Si}$  before and after cycling

For the  $\text{Li}_2\text{S}/\text{graphene}$  nanocomposite cathode, scanning electron microscope (SEM) and XRD were used to characterize their composition and structure, as shown in Fig. 3a, b. The XRD pattern (Fig. 3a) reveals the presence of only the pure and crystalline phase of  $\text{Li}_2\text{S}$  without any impurities (JCPDS card no. 26-1188), which suggests no chemical reactions occur between  $\text{Li}_2\text{S}$ , graphene, and anhydrous ethanol. The SEM-EDS elemental mapping in Fig. 3b shows that the element S is homogeneously distributed on the surface of the graphene. For  $\text{Li}_2\text{S}/\text{G}$  composites, the enhanced electronic conductivity and improved active material utilization can contribute to its enhanced electrochemical performance. The voltage profile (Fig. 3c) shows that the first charge platform is much different from that of subsequent charges. The first charge voltage is higher up to  $\sim 2.8$  V, while the voltage hysteresis of the subsequent charge–discharge becomes significantly smaller. It might be attributed to the fact that the initial extraction of Li from crystalline  $\text{Li}_2\text{S}$  requires higher overpotential, which has been reported by many published research results [34–36]. Figure 3d shows the discharge capacity over a number of cycles of the all-solid-state battery with  $\text{Li}_2\text{S}$  cathode. The initial discharge capacity is 719.2 mAh/g, and the capacity retains 371.2 mAh/g after 20 cycles. The low  $\text{Li}_2\text{S}$  utilization probably result from the fact that  $\text{Li}_2\text{S}$  particle needs a direct contact with electronic/ionic conductive additives for electron and  $\text{Li}^+$  transfer during electrochemical cycling, which is difficult to realize in the all-solid-state battery. The incomplete contact between  $\text{Li}_2\text{S}$  and electronic/ionic conductive additives would decrease the active material utilization, causing a rather low capacity and Coulomb efficiency [35, 37]. Besides, the low ionic conductivity of the solid electrolyte and the large interfacial resistance between electrolyte and electrode in the all-solid-state batteries also have a negative effect on the  $\text{Li}_2\text{S}$  utilization [36].

The main purpose of our work is to incorporate  $\text{Li}_2\text{S}$  instead of sulfur as the cathode with a Si counter electrode, which could avoid the interface problems and obtain a battery system with the high energy density. The  $\text{Si@LPS}$  and  $\text{Li}_2\text{S}/\text{G}$  were characterized and tested in all-solid-state half-cells, just

as mentioned above. After that, all-solid-state  $\text{Li}_2\text{S}/\text{Si}$  full cells were fabricated for the electrochemical tests. As displayed in Fig. 4a, b, the voltage profile shows a large voltage difference in the charge and discharge voltage platform, which is caused by the severe polarization potential for the all-solid-state battery [35, 36, 38]. The curves for the all-solid-state battery show only one platform of charge and discharge, which is in accordance with the CV results (Fig. S1). The full cell has an initial discharge capacity of 316.9 mAh/g, and it retains 179.4 mAh/g after 20 cycles. From comparison of Figs. 3d and 4b, we can see that the specific capacity of the  $\text{Li}_2\text{S}/\text{Si}$  all-solid-state full cell decays faster than that of the  $\text{Li}_2\text{S}/\text{Li}$  half-cell. It is probably ascribed to the fact that there is a limited supply of Li ions in the full cells of  $\text{Li}_2\text{S}/\text{Si}$ , which can be irreversibly lost in side reactions during cycling process [34, 36].

The electrochemical impedance spectroscopy measurement (EIS) of all-solid-state cells before and after cycling was executed, as presented in Fig. 5. The impedance spectra show that the internal resistance of the all-solid-state half-batteries based on  $\text{Si@LPS}$  or  $\text{Li}_2\text{S}/\text{G}$  increases largely after 20 cycles due to the interface reaction between LPS and lithium metal. However, the internal resistance of the all-solid-state full battery is still small, as shown in Fig. 5c. It indicates that the  $\text{Si}/\text{Li}_2\text{S}$  has a more stable interface between electrode and LPS solid electrolyte [39, 41]. We believe that better cycling behavior and capacity retention can be obtained through optimization of the all-solid-state  $\text{Li}_2\text{S}/\text{Si}$  system, including electrode optimization and interface modification.

## Conclusion

In conclusion, we report an all-solid-state full battery system with  $\text{Li}_2\text{S}$  cathode, Si anode, and LPS electrolyte. Si nanoparticles are in-suit coated with LPS, which could provide an intimate interfacial contact, suppress Si volume expansion, and promote  $\text{Li}^+$  diffusion during cycling process.  $\text{Li}_2\text{S}/\text{G}$  is used as lithium source, overcoming the interface problems between lithium metal and electrolyte. With a higher ionic

conductivity solid electrolyte, the  $\text{Li}_2\text{S}/\text{Si}$  full battery shows a relatively impressing electrochemical performance at room temperature. This all-solid-state battery system we propose could avoid the safety issues associated with the use of lithium metal and have a potential for an enhanced energy density, which would have a significant impact on their applications in the fields of electric vehicles and portable electronics. Further investigation will be carried out to further improve the battery cycling behavior by optimizing the electrodes and modifying the interfaces between the solid-state electrolyte and the electrodes.

**Funding information** The authors acknowledge funding support from 1000 Talent Plan program (NO. 31370086963030), research projects from Shandong Province (2018JMRH0211, 2017CXGC1010, and 2016GGX104001), Taishan Scholar Program (11370085961006), the National Science Foundation of Shandong Province (ZR2017MEM002), and the Fundamental Research Funds of Shandong University (201810422046, 2017JC010, 2017JC042, and 2016JC005).

## References

- Falco M, Castro L, Nair J, Bella F, Bardé F, Meligrana G, Gerbaldi C (2019) UV-cross-linked composite polymer electrolyte for high-rate, ambient temperature lithium batteries. *ACS Applied Energy Mater* 2(3):1600–1607
- Nair R, Colò F, Kazzazi A, Moreno M, Bresser D, Lin R, Bella F, Meligrana G, Fantin S, Simonetti E, Appetecchi G, Passerini S, Gerbaldi C, Appetecchi B (2019) Room temperature ionic liquid (RTIL)-based electrolyte cocktails for safe, high working potential Li-based polymer batteries. *J Power Sources* 412:398–407
- Radzir M, Hanifah S, Ahmad A, Hassan N, Bella F (2015) Effect of lithium bis (trifluoromethylsulfonyl) imide salt-doped UV-cured glycidyl methacrylate. *J Solid State Electrochem* 19(10):3079–3085
- Yang Q, Li W, Dong C, Ma Y, Yin Y, Wu Q, Xu Z, Ma W, Fan C, Sun K (2020) PIM-1 as an artificial solid electrolyte interphase for stable lithium metal anode in high-performance batteries. *J Energy Chem* 42:83–90
- Kwak W, Park J (2019) Cathode coating using  $\text{LiInO}_2\text{-LiI}$  composite for stable sulfide-based all-solid-state batteries. *Sci Rep* 9(1):8099
- Ma J, Chen B, Wang L, Wang L, Cui G (2018) Progress and prospect on failure mechanisms of solid-state lithium batteries. *J Power Sources* 392:94–115
- Zhang J, Zhao J, Yue L, Wang Q, Chai J, Liu Z, Zhou X, Li H, Guo Y, Cui G, Chen L (2015) Safety-reinforced poly (propylene carbonate)-based All-solid-state polymer electrolyte for ambient-temperature solid polymer lithium batteries. *Adv Energy Mater* 5(24):1501082
- Zhang J, Zheng C, Lou J, Xia Y, Liang C, Huang H, Zhang W (2019) Poly (ethylene oxide) reinforced  $\text{Li}_6\text{PS}_5\text{Cl}$  composite solid electrolyte for all-solid-state lithium battery: enhanced electrochemical performance, mechanical property and interfacial stability. *J Power Sources* 412:78–85
- Wan Z, Lei D, Yang W, Liu C, Shi K, Hao X, Kang F (2019) All-solid-state batteries: low resistance-integrated all-solid-state battery achieved by  $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$  nanowire upgrading polyethylene oxide (PEO) composite electrolyte and PEO cathode binder. *Adv Funct Mater* 29(1):1970006
- Yu X, Xue L, Goodenough JB, Manthiram A (2019) A high-performance all-solid-state sodium battery with a poly (ethylene oxide)- $\text{Na}_3\text{Zr}_2\text{Si}_2\text{PO}_{12}$  composite electrolyte. *ACS Mater Lett* 1:132–138
- Anastro A, Lago N, Berlanga C, Galcerán M, Hilder M, Forsyth M, Mecerreyes D (2019) Poly (ionic liquid) iongel membranes for all solid-state rechargeable sodium battery. *J Membr Sci* 582:435–441
- Miura A, Rosero-Navarro NC, Sakuda A, Tadanaga K, Phuc N, Matsuda A, Tatsumisago M (2019) Liquid-phase syntheses of sulfide electrolytes for all-solid-state lithium battery. *Nat Rev Chem* 3:189–198
- Kim K, Park J, Jeong G, Yu J, Kim Y, Park S, Kanno R (2019) Rational design of a composite electrode to realize a high-performance all-solid-state battery. *ChemSusChem* 12:2637–2643
- Ates T, Keller M, Kulisch J, Adermann T, Passerini S (2019) Development of an all-solid-state lithium battery by slurry-coating procedures using a sulfidic electrolyte. *Energy Storage Mater* 17:204–210
- Qu H, Ju J, Chen B, Xue N, Du H, Han X, Zhang J, Xu G, Yu Z, Wang X, Cui G (2018) Inorganic separators enable significantly suppressed polysulfide shuttling in high-performance lithium-sulfur batteries. *J Mater Chem A* 6(46):23720–23729
- Suriyakumar S, Gopi S, Kathiresan M, Bose S, Gowd B, Nair R, Bose S, Angulakshmi N, Meligrana G, Gerbaldi G, Stephan M (2018) Metal organic framework laden poly (ethylene oxide) based composite electrolytes for all-solid-state Li-S and Li-metal polymer batteries. *Electrochim Acta* 285:355–364
- Pan H, Huang X, Zhang R, Zhang T, Chen Y, Hoang T, Wen G (2018) Reduced graphene oxide-encapsulated mesoporous silica as sulfur host for lithium-sulfur battery. *J Solid State Electrochem* 22(11):3557–3568
- Li L, Zhong B (2018) The design and preparation of the composite with layered spherical structure for Li-S battery. *J Solid State Electrochem* 22(2):591–598
- Nara H, Tsuda S, Osaka T (2017) Techniques for realizing practical application of sulfur cathodes in future Li-ion batteries. *J Solid State Electrochem* 21(7):1925–1937
- Sun C, Liu J, Gong Y, Wilkinson DP, Zhang J (2017) Recent advances in all-solid-state rechargeable lithium batteries. *Nano Energy* 33:363–386
- Tian YS, Shi T, Richards W, Li JC, Kim J, Bo SH, Ceder G (2017) Compatibility issues between electrodes and electrolytes in solid-state batteries. *Energy Environ Sci* 10(5):1150–1166
- Chen B, Ju J, Ma J, Zhang J, Xia R, Cui G, Chen L (2017) An insight into intrinsic interfacial properties between Li metals and  $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$  solid electrolytes. *Phys Chem Chem Phys* 19(46):31436–31442
- Zhong C, Deng Y, Hu W, Qiao L, Zhang L, Zhang J (2015) A review of electrolyte materials and compositions for electrochemical supercapacitors. *Chem Soc Rev* 44(21):7484–7539
- Manthiram A, Yu X, Wang S (2017) Lithium battery chemistries enabled by solid-state electrolytes. *Nature Rev Mater* 2(4):16103
- Chien P, Feng X, Tang M, Rosenberg JT, Neill SO, Zheng J, Grant S, Hu Y (2018) Li Distribution heterogeneity in solid electrolyte  $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$  upon electrochemical cycling probed by  $^7\text{Li}$  MRI. *J Phys Chem Lett* 9(8):1990–1998
- Ban X, Zhang W, Chen N, Sun C (2018) A high-performance and durable poly (ethylene oxide)-based composite solid electrolyte for all solid-state lithium battery. *J Phys Chem C* 122(18):9852–9858
- Zhang J, Zang X, Wen H, Dong T, Chai J, Li Y, Chen B, Zhao J, Dong S, Ma J, Yue L, Liu Z, Guo X, Cui G, Chen L (2017) High-voltage and free-standing poly (propylene carbonate)/ $\text{Li}_6.75\text{La}_3\text{Zr}_{1.75}\text{Ta}_{0.25}\text{O}_{12}$  composite solid electrolyte for wide temperature range and flexible solid lithium ion battery. *J Mater Chem A* 5(10):4940–4948

28. Wenzel S, Randau S, Leichtweiss T, Weber DA, Sann J, Zeier WG, Janek J (2016) Direct observation of the interfacial instability of the fast-ionic conductor Li<sub>10</sub>GeP<sub>2</sub>S<sub>12</sub> at the lithium metal anode. *Chem Mater* 28(7):2400–2407
29. Hayashi A, Ohtsubo R, Ohtomo T, Mizuno F, Tatsumisago M (2016) All-solid-state rechargeable lithium batteries with Li<sub>2</sub>S as a positive electrode material. *J Power Sources* 183(1):422–426
30. Shin B, Nam YJ, Oh DY, Kim DH, Kim J, Jung Y (2014) Comparative study of TiS<sub>2</sub>/Li-In all-solid-state lithium batteries using glass-ceramic Li<sub>3</sub>PS<sub>4</sub> and Li<sub>10</sub>GeP<sub>2</sub>S<sub>12</sub> solid electrolytes. *Electrochim Acta* 146:395–402
31. Zhang Z, Zhao Y, Chen S, Xie D, Yao X, Cui P, Xu X (2017) An advanced construction strategy of all-solid-state lithium batteries with excellent interfacial compatibility and ultralong cycle life. *J Mater Chem A* 5(32):16984–16993
32. Xu X, Hou G, Nie X, Ai Q, Liu Y, Feng J, Zhang L, Si P, Guo S, Ci L (2018) Li<sub>7</sub>P<sub>3</sub>S<sub>11</sub>/poly (ethylene oxide) hybrid solid electrolytes with excellent interfacial compatibility for all-solid-state batteries. *J Power Sources* 400:212–217
33. Xu X, Ai Q, Pan L, Ma X, Zhai W, An Y, Hou G, Chen J, Zhang L, Si P, Lou J, Feng J, Ci L (2018) Li<sub>7</sub>P<sub>3</sub>S<sub>11</sub> solid electrolyte coating silicon for high-performance lithium-ion batteries. *Electrochim Acta* 276:325–332
34. Yao X, Liu D, Wang C, Long P, Peng G, Hu Y, Li H, Chen L, Xu X (2016) High-energy all-solid-state lithium batteries with ultralong cycle life. *Nano Lett* 16(11):7148–7154
35. Yang Y, McDowell MT, Jackson M, Cha JJ, Hong SS, Cui Y (2010) New nanostructured Li<sub>2</sub>S/silicon rechargeable battery with high specific energy. *Nano Lett* 10(4):1486–1491
36. Lin Z, Liu Z, Dudney N, Liang C (2013) Lithium superionic sulfide cathode for all-solid lithium–sulfur batteries. *ACS Nano* 7(3):2829–2833
37. Wu FW, Kim H, Magasinski A, Lee JT, Lin HT, Yushin G (2014) Harnessing steric separation of freshly nucleated Li<sub>2</sub>S nanoparticles for bottom-up assembly of high-performance cathodes for lithium–sulfur and lithium-ion batteries. *Adv Energy Mater* 4(11):1400196
38. Cai K, Song M, Cairns E, Zhang Y (2012) Nanostructured Li<sub>2</sub>S–C composites as cathode material for high-energy lithium/sulfur batteries. *Nano Lett* 12(12):6474–6479
39. Yang T, Wang X, Wang D (2016) Facile and scalable synthesis of nanosized core–shell Li<sub>2</sub>S@ C composite for high-performance lithium–sulfur batteries. *J Mater Chem A* 4(42):16653–16660
40. Yu M, Wang Z, Wang Y, Dong Y, Qiu J (2017) Freestanding flexible Li<sub>2</sub>S paper electrode with high mass and capacity loading for high-energy Li–S batteries. *Adv Energy Mater* 7(17):1700018
41. Wan H, Pierre J, Qi X, Liu X, Xu X, Li H, Hu Y, Yao Y (2018) Core–Shell Fe<sub>1–x</sub>S@Na<sub>2.9</sub>PS<sub>3.95</sub>Se<sub>0.05</sub> nanorods for room temperature all-solid-state sodium batteries with high energy density. *ACS Nano* 12(3):2809–2817

**Publisher's note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.